



Society of Petroleum Engineers

SPE-229390-MS

Smart Liquid Unloading IIoT Application for Gas Wells in the Haynesville Basin

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229390-MS](https://doi.org/10.2118/229390-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Intermittent wells, particularly in gas-producing basins like Haynesville, are traditionally managed with manual or partially automated controls, requiring frequent operator intervention and relying on static SCADA infrastructure. This approach limits production efficiency and responsiveness to changing well conditions, especially in the presence of liquid loading. To address these limitations, this paper presents an autonomous, data-driven solution deployed at the edge, designed specifically for intermittent well optimization.

The proposed system is an IIoT-enabled application running on a local edge gateway. It leverages real-time sensor data—such as casing and tubing pressures, choke position, flow rates, and temperature—to dynamically calculate key flow parameters including liquid level and critical velocity. These parameters are used to continuously evaluate well conditions and automatically adjust operating setpoints in a closed-loop manner. The result is a self-adaptive control system capable of minimizing downtime and maximizing production without human intervention.

This application was deployed on multiple wells across eight pads in the Haynesville basin. Despite variability in well conditions from long cycles to short and quick, the solution delivered consistently strong results. Over optimization periods ranging from 63 to 83 days, wells showed improvements in production between 70% and 139% in cumulative gas output, with average daily rates up to 350 MCFD. Economic analysis estimates an additional 80 MMCF of gas production per year per well, translating to over \$135,000 in added value, with a modest yearly investment per well of approximately \$3,600.

This study demonstrates how edge-based, autonomous optimization modernizes intermittent well management by enhancing liquid unloading efficiency and unlocking measurable production and economic gains. It presents a scalable deployment model supported by a structured field personnel training framework to accelerate full-field adoption. Critically, in environments with basic remote infrastructure—such as actuated chokes or valves—this solution can be implemented with minimal additional capital investment, enabling rapid, cost-effective deployment across large well populations.

Introduction

Gas wells transitioning into intermittent production often face significant operational inefficiencies due to liquid loading, especially in high-pressure basins such as the Haynesville basin. In these later-life wells, bottomhole pressure becomes insufficient to continuously lift liquids, causing flow interruptions and pressure buildup. Managing these wells typically involves manually timed shut-ins or basic SCADA-triggered control logic, both of which are reactive, labor-intensive, and unable to adapt to evolving well conditions.

This challenge is exacerbated by legacy infrastructure: intermittent wells often rely on fixed logic based on casing-to-tubing pressure ratios or calendar-based cycling, without continuous feedback from dynamic flow conditions. As a result, these wells frequently experience suboptimal unloading, prolonged downtime, and lost production. Operators are also constrained by the limited responsiveness of centralized SCADA systems and the need for frequent field visits to adjust choke settings or validate production trends.

To address these limitations, this paper introduces a field-proven Industrial Internet of Things (IIoT) application that autonomously manages the liquid unloading cycle using a data-driven, edge-based control system. Running directly on a local gateway device at the well site, the application continuously ingests real-time pressure, flowrate, and temperature data. It applies embedded physics-based models and lightweight machine learning algorithms to calculate key flow parameters and dynamically adjusts choke settings in a closed-loop manner without requiring cloud connectivity or operator intervention.

While the act of unloading a well may appear operationally simple, optimizing the process in real time is far more complex. It requires integrating multiple dynamic variables, such as pressure buildup trends, gas velocity thresholds, choke actuation behavior, liquid holdup, temperature gradients, and wellbore geometry, all of which interact nonlinearly and evolve continuously. Attempting to model these interactions deterministically is both technically demanding and operationally impractical without the aid of intelligent, automated systems.

In addition to technical hurdles, intermittent wells also face unique economic constraints. These are typically late-life assets, operating under tighter capital budgets. Traditional optimization approaches, such as retrofitting wells with advanced sensors or implementing custom automation platforms, often require significant upfront cost and engineering time. These investments are difficult to justify at scale across large populations of marginal wells. In this context, data-driven solutions become not only a technical enabler but a financial necessity, they allow for cost-effective optimization using existing infrastructure, reducing the need for capital-heavy upgrades while still unlocking meaningful production gains.

Building on the architecture successfully deployed in artificial lift applications such as autonomous SRP optimization frameworks that leverage local intelligence to adapt rod-pump control logic without cloud reliance ([Gambaretto, et al., 2024](#)), this solution demonstrates how real-time control logic can be extended to intermittent gas wells to maximize production, minimize downtime, and enable scalable automation. Similar data-driven approaches have shown strong performance in gas-lift optimization scenarios, where edge-deployed models adapt to variable flow regimes with minimal human tuning ([Gambaretto & Kashif, 2023](#)).

Methodology

Architecture Overview

The smart unloading solution is built around an edge-deployed application running on a ruggedized field gateway. This device integrates directly with existing wellhead instrumentation (casing pressure, tubing pressure, temperature, flowrate, and choke actuator) requiring no modifications to upstream SCADA or communications infrastructure.

The edge application is composed of four tightly integrated layers:

- Data Acquisition Layer: Continuously collects and validates telemetry from the wellhead, including pressure, temperature, and flowrate sensors.
- Calculation Engine: This module computes flow parameters essential to understanding unloading behavior. It applies embedded physics-based models to derive:
 - Gas velocity, using surface flowrate and tubing geometry
 - Critical velocity, based on Turner's mechanistic model for multiphase flow, defines the minimum gas velocity needed to lift liquids. The framework is compatible with alternative models (e.g., Coleman, Nosseir, Barnea) to support different flow regimes and tubing profiles
 - Liquid column height, inferred from differential casing-to-tubing pressures and adjusted for gas compressibility and temperature

These calculations are continuously updated to reflect changing wellbore conditions. As flow rate declines, the system compares real-time gas velocity to the dynamic critical threshold to assess whether liquid is still being effectively carried.

- Autonomous Decision Logic: The core control algorithm replaces static calendar-based cycling with real-time, condition-driven logic. It uses calculated parameters to autonomously decide when to shut in the well. Specifically, it monitors:
 - Gas velocity versus critical velocity thresholds
 - Estimated liquid level and liquid accumulation rate

When gas velocity drops below the calculated critical velocity and the inferred liquid column reaches a calculated tipping point, the system preemptively shuts in the well. This prevents the well from slipping into a deeply loaded state that would require extended shut-in or operator intervention such as flowing the well to a tank causing unwanted greenhouse gas emissions. Conversely, the system avoids premature shut-in by recognizing short-lived flow fluctuations and only acting when degradation is consistent across multiple time windows. This adaptive mechanism replaces calendar-based cycling with condition-driven control.

- ML-Based Shut-In Duration Estimation: Instead of applying a fixed shut-in time, the duration is dynamically computed by a machine learning (ML) model that could be deployed at the edge, however, to enhance scalability and agility, it was moved to the cloud. The model predicts the optimal shut-in duration to allow the required pressure recharge to resume effective flow, taking into account:
 - Historical shut-in pressure trends
 - Prior cycle efficiency and duration
 - Real-time pressure build-up behavior
 - Well-specific flowback performance

This ensures that the well is reopened at the precise moment when pressure is sufficient to unload without waiting too long (which would waste valuable production time) or restarting too early (which could result in failed or weak unloading). In practice, the ML model uses both pressure buildup curves and prior flowback success metrics to forecast the optimal shut-in duration. For example, if a well consistently underperforms when reopened after 6 hours but shows strong recovery at 8 hours, the model adjusts future shut-ins accordingly. This adaptive learning process is implemented independently for each well, leveraging Long-Short Term Memory (LSTM)-based

time-series models that capture both seasonality and trend shifts in well behavior. Retraining occurs periodically as new cycles are completed, enabling the system to evolve with changing reservoir and equipment conditions

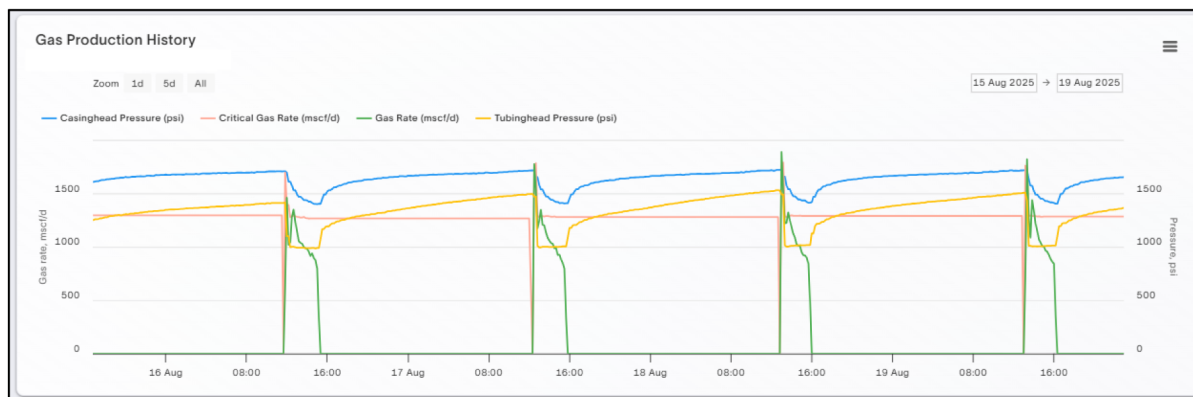


Figure 1—Well shut-in base Critical Gas Rate vs Real Gas Rate and loading start point

By combining deterministic physics with ML-driven timing, the system achieves an adaptive, intelligent cycle management strategy that maximizes gas recovery while minimizing manual intervention.

Machine Learning Workflow for Adaptive Shut-In Optimization

The foundation of any machine learning model is data. That's why the success of ML depends on acquiring the data, shaping it into the required format, selecting essential features and generating new features if necessary. While the edge device easily manages the first step, the raw field data typically comes with substantial noise, especially in the high-frequency range. Following practices similar to those used in prior smart valve optimization work (Chen, et al., 2024), we segmented the dataset into distinct operational cycles, each beginning with a shut-in phase and continuing through the subsequent flow period. This cycle-based structuring improved model performance by isolating pressure build-up behavior and mitigating the impact of high-frequency noise on prediction accuracy. Once cycles are identified the algorithm evaluates the success of cycles, only successful cycles data are used for the training. The first model to predict the pressure build-up is trained on the extracted pressures values during the well close state.

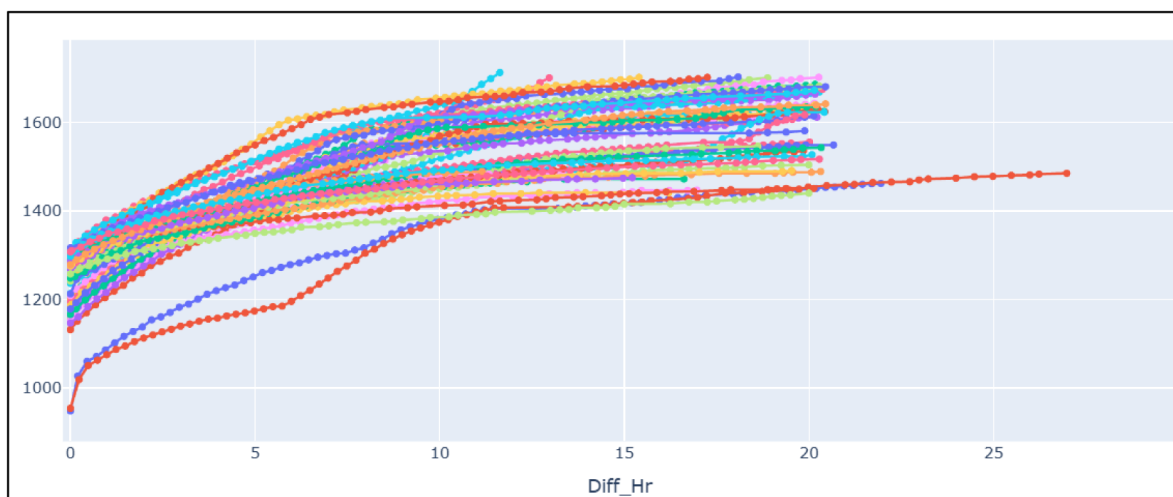


Figure 2—THP build up after removing noise for one well

The second model is using statistics per cycle (Figure 3–4). We have a model per well and for every well the training is implemented automatically by evaluating multiple regression algorithms, the best fitting one is used to predict the production rate for the following open state based on the forecasted pressure values and the variable "Shut-in duration, hr.", which allow us to generate the curve (Figure 5) to find the best timing to open the well to maximize production.

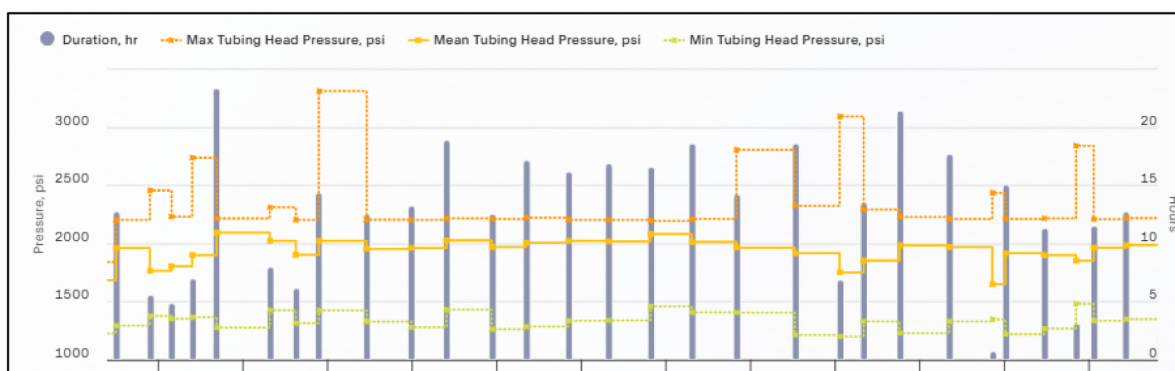


Figure 3—Well Close state statistics overtime

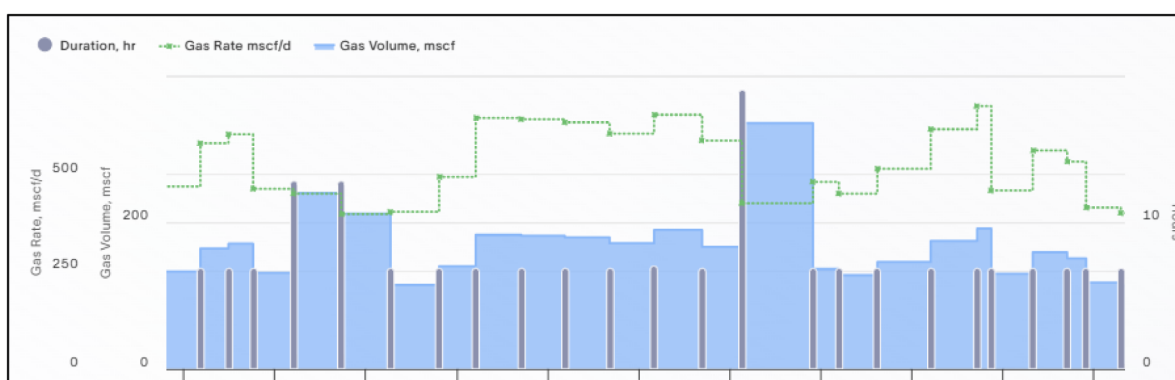


Figure 4—Well Open state statistics overtime

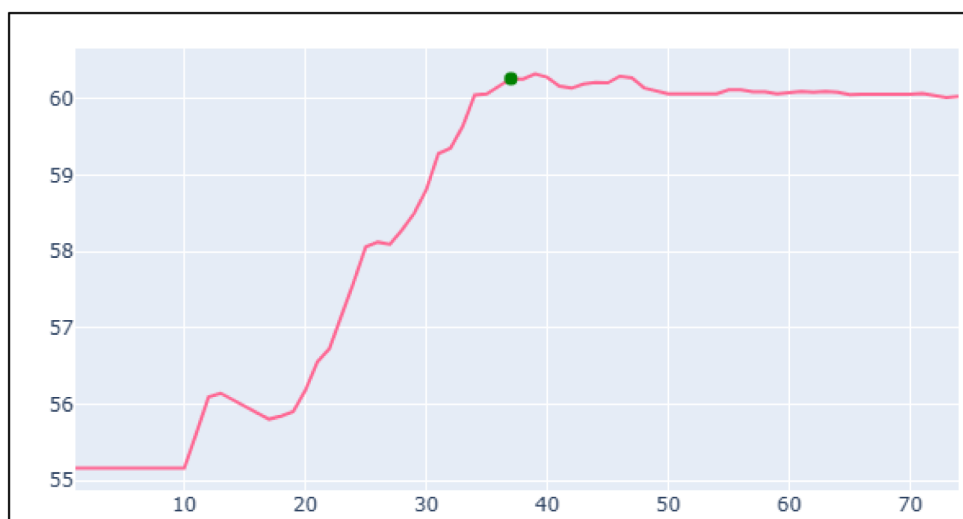


Figure 5—Average production rate vs Shut-in time, hr.

As illustrated in Figure 5, a shut-in duration of approximately 37 hours (green dot) optimizes the average production during the subsequent open period. Beyond this threshold, production rates do not exhibit further improvement, indicating that longer shut-in times are not advantageous.

ML Ops Integration via Cloud for Model Retraining

While the real-time decisions for well control are executed at the edge, the machine learning (ML) models that support shut-in duration optimization are hosted and maintained in the cloud to favor scalability. This architecture balances the need for fast local action with the flexibility and maintainability of centralized model training.

Operational Workflow. The overall process works as follows:

1. **Autonomous Edge Control:** The edge device ingests real-time sensor data (pressures, flow rate, temperature) and autonomously makes the decision to shut in the well based on current conditions.
2. **Data Capture for ML Inference:** Once the valve is closed, the edge system records the initial pressure buildup behavior (typically the first few minutes after shut-in) along with relevant metadata from the preceding production cycle.
3. **Cloud-Based ML Inference:** This cycle summary is then transmitted to the cloud where the ML model evaluates the pressure buildup trend and forecasts the optimal shut-in duration for that particular well and cycle. The output (the recommended shut-in time) is returned to the edge gateway as input for the ongoing cycle.
4. **Cycle Execution:** The edge system uses this prediction to autonomously reopen the valve at the calculated optimal time balancing sufficient pressure recharge with the risk of over-shut-in and lost production.
5. **Model Maintenance and Retraining (ML Ops):** Storing and processing ML models in the cloud simplifies retraining workflows. As more cycles accumulate and conditions evolve (e.g., changes in reservoir pressure, liquid loading behavior, or equipment wear), the cloud-hosted models are periodically retrained with updated data. This avoids the logistical complexity of deploying retrained models to distributed edge devices, making model lifecycle management (ML Ops) significantly more efficient.

Advantages of Cloud-Based ML Hosting.

- **Scalability and agility:** Model training, validation, and deployment are centralized, making it easier to apply consistent methodologies across large well populations.
- **Maintainability:** New training pipelines, feature engineering techniques, or algorithm upgrades can be rolled out without firmware updates or edge device access.
- **Adaptability:** Models are retrained continuously or on-demand as new production data becomes available, improving long-term prediction accuracy.

Cloud Integration and Operational Oversight

In addition to autonomous edge execution, the system includes a cloud-based component that enables remote monitoring, historical analysis, and management by exemption dashboards. While all cycle-level decisions are made locally at the wellsite, the cloud environment collects summarized data across multiple wells to support operational oversight and engineering review.

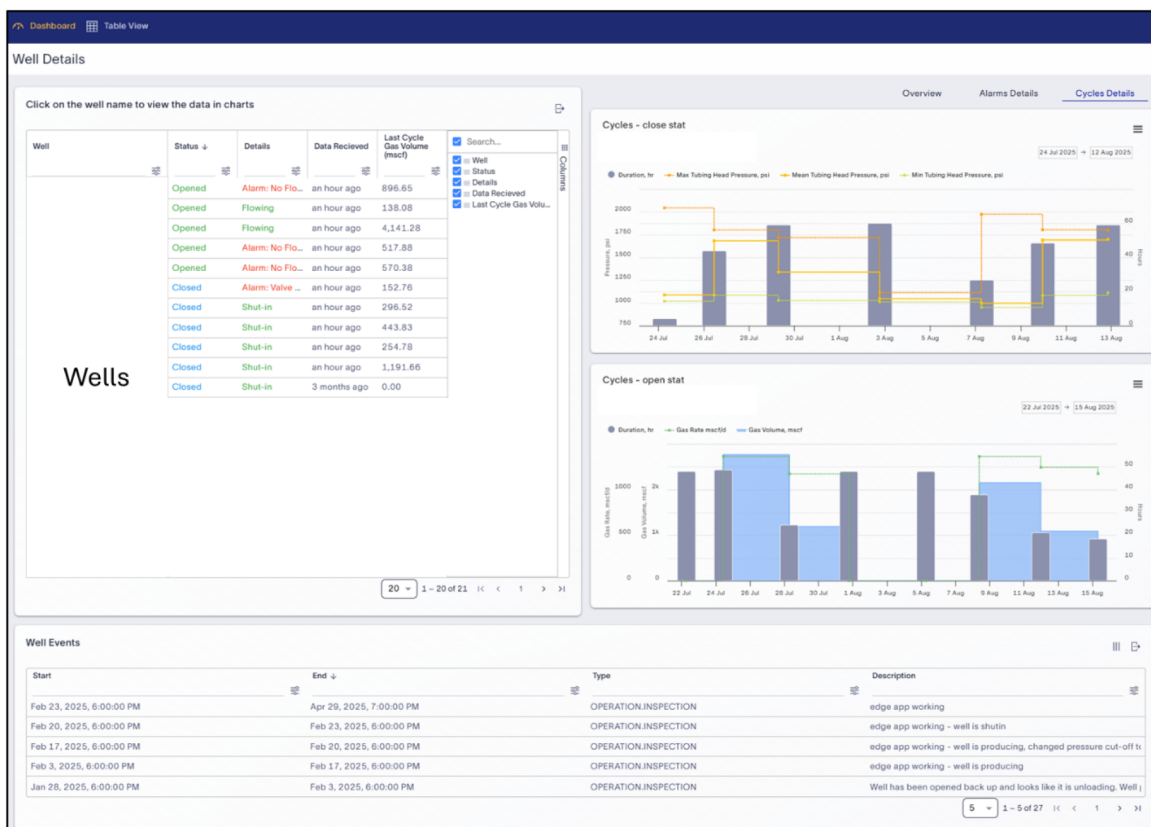


Figure 6—Landing dashboards

Lightweight Data Transmission. To reduce transmission costs and avoid overloading network infrastructure, only aggregated data is sent to the cloud. This includes metadata such as shut-in durations, gas flowback rates, pressure buildup profiles, and ML outputs. Full resolution time series are retained locally, and only key summaries or anomalies are transmitted. This design minimizes cellular or satellite bandwidth usage, which is especially important in remote areas.

This data-throttling strategy ensures that the edge device remains self-sufficient and continues operation even in low-connectivity environments, while still allowing centralized visibility for engineering teams.

Alarm-Based Field Intervention in Autonomous Operations. Because the edge application is designed to operate fully autonomously, executing liquid unloading cycles without manual input, it is essential to include robust alarm-based safeguards to catch anomalies and trigger field attention when necessary. These alarms ensure reliable operations and operational continuity in decentralized environments. The alarm suite currently includes:

- **Valve Non-Responsive Alarm:** Triggered when the valve does not respond within a specified time after a command is issued. This could indicate a hardware fault or communication loss.
- **Valve Leakage Alarm:** Raised when flow is detected while the valve is closed indicating internal leakage or mechanical failure.
- **No-Flow-on-Open Alarm:** Activated when the valve is open but the well shows no flow, which may point to production impairment, backpressure issues, or valve not opening.

These alarms are continuously monitored at the edge level, with relevant alerts and metadata forwarded to the cloud platform, where engineers are notified only when attention is truly needed.

Management by Exception. This alarm system directly supports management by exception model. Instead of engineers manually monitoring each well or reviewing performance data for hundreds of assets, they

are automatically alerted only when abnormal conditions are detected. This significantly reduces routine workload while ensuring that critical issues are addressed in a timely manner. Management by exception ensures that engineering resources are focused where they matter most.

Wells

Select the well name to see more details.

Entity Name	State	Details	Data Received	App Active	Valve Response Alarm	Well Flow Alarm	Tubing Head Pressure (psig)	THP Cutoff (psig)	Gas Rate (Mscfd)	Gas Flow Rate Cutoff (Mscfd)
Wells	Open	Flowing	3 hours ago	●	●	●	2,292.00	2,200.00	289.00	400.00
	Open	Alarm: No Flow	3 hours ago	●	●	●	1,001.00	1,600.00	0.00	600.00
	Open	Flowing	3 hours ago	●	●	●	991.00	2,000.00	1,439.00	800.00
	Open	Alarm: No Flow	3 hours ago	●	●	●	1,756.00	1,700.00	0.00	500.00
	Open	Alarm: No Flow	3 hours ago	●	●	●	1,706.00	1,600.00	0.00	300.00
	Closed	Shut-in	3 hours ago	●	●	●	1,296.00	1,700.00	0.00	400.00
	Closed	Shut-in	3 hours ago	●	●	●	1,726.00	1,750.00	1,224.00	200.00
	Closed	Shut-in	3 hours ago	●	●	●	1,571.00	1,575.00	0.00	600.00
	Closed	Shut-in	3 hours ago	●	●	●	1,545.00	1,800.00	0.00	300.00
	Closed	Shut-in	3 hours ago	●	●	●	1,366.00	1,430.00	0.00	500.00
	Closed	Shut-in	3 months ago	●	●	●	1,976.00	2,000.00	0.00	800.00

Figure 7—Management by exemption dashboard

By combining autonomous control with targeted exception monitoring, this architecture achieves scalable, cost-efficient field management without compromising safety or visibility.

Configurable Control Parameters. Cloud access allows centralized configuration management. Engineers can define per-well constraints such as:

- Minimum and maximum shut-in durations
- Maximum and minimum open durations
- Fallback behavior (e.g., revert to pressure threshold logic or fixed cycling)
- Gas production rate thresholds for closing

These constraints act as safety boundaries that prevent anomalous or unsafe cycling behavior due to sensor faults, outlier model predictions, or unexpected well conditions. The system also supports hybrid modes where ML logic is supplemented by physical rules or operator-defined setpoints.

Operational and Financial Impact. This architecture minimizes total cost of ownership by avoiding centralized compute dependency, reducing network transmission expenses, and scaling efficiently across assets.

Key benefits include:

- Reduced bandwidth usage through selective data publishing
- Lower communication costs in satellite or cellular environments
- Minimal infrastructure overhead—no full-time cloud compute required
- Autonomous failover: wells continue optimizing even without connectivity
- Scalable fleet-wide oversight using lightweight, exception-based dashboards

The cloud layer complements edge autonomy with a centralized view, actionable alerts, and collaboration tools—making it a critical component for field-wide deployment and sustainable scaling.

Field Implementation and Results

Deployment Scope

The autonomous unloading solution was deployed across several pads and piloted in 9 wells in the Haynesville Basin, covering a range of gas wells under intermittent production. These assets presented diverse production behaviors, including:

- Long shut-in durations followed by rapid flowback
- Shorter, more frequent unloading cycles
- Varying degrees of liquid loading severity

Candidate wells were selected based on current production behavior and infrastructure availability. Key selection criteria included:

- the presence of RTU with communication capability,
- remote-actuated chokes,
- reliable cell service,
- solar power capacity and
- no need for major mechanical repairs or modifications.

The containerized application was installed on IIoT gateways at each pad and grouping as many well as possible under individual devices, requiring minimal hardware connections or SCADA modifications. All wells had basic instrumentation (casing pressure, tubing pressure and temperature, and flowmeter data) plus remote-actuated chokes.

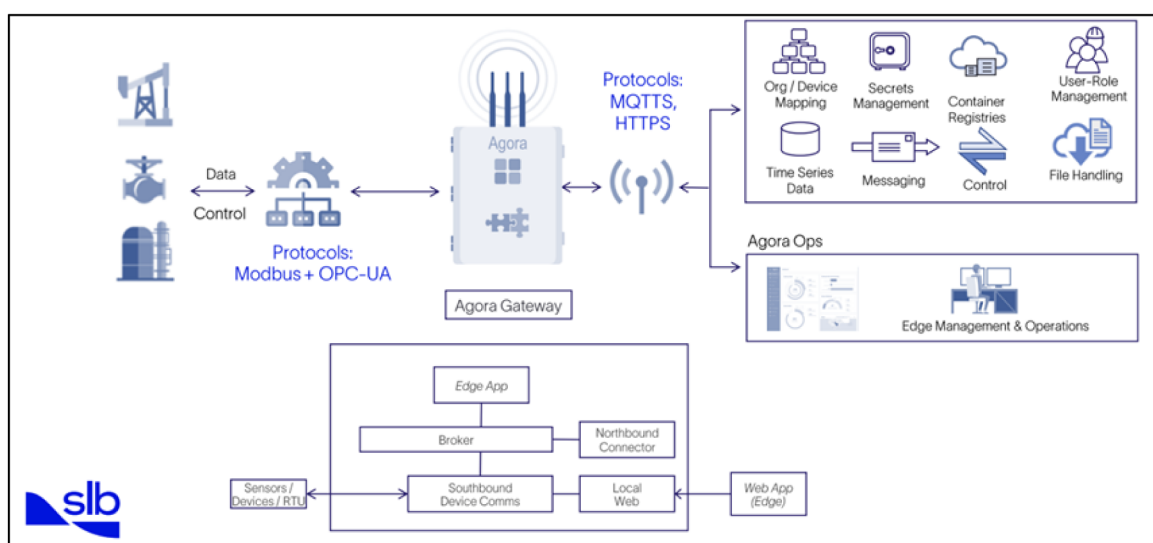


Figure 8—Edge architecture

Production Outcomes

The autonomous system demonstrated substantial production improvements over the 63–83 days of optimization periods.

- Cumulative production increased by 70% to 139% compared to historical baselines

- Daily production gains reached up to 350 MCFD in high-potential wells
- Estimated over 80 MMCF additional gas produced over 1 year per well
- Per-well value creation exceeding \$135,000 annually

Although this methodology was new to the field personnel during the pilot, certain well checks were still required. Despite this, the objectives were successfully achieved while maintaining fully autonomous operations, eliminating the need for manual choke cycling or shut-in scheduling by operators.

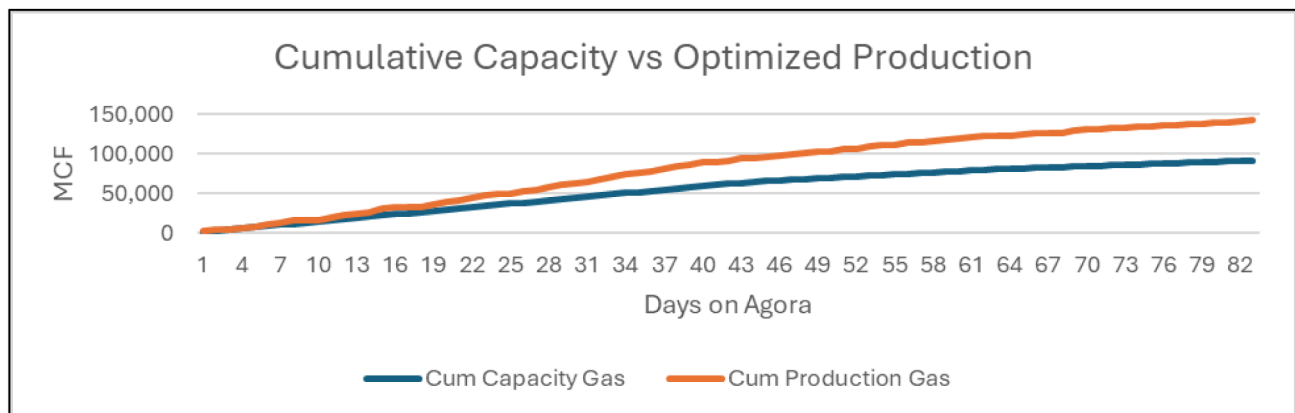


Figure 9—Production baseline vs. optimized

Table 1—Production Results per Well

Well	Days Since Start	Cumulative Capacity (MCF)	Cumulative Production (MCF)	% Change
Well 1	63	5,864.20	14,012.70	139.0%
Well 2	83	10,401.79	22,338.04	114.8%
Well 3	83	14,533.86	24,807.80	70.7%
Well 4	43	4,870.73	7,999.39	64.2%
Well 5	43	7,412.47	10,885.19	46.8%
Well 6	48	3,938.57	5,697.60	44.7%
Well 7	43	8,914.34	12,213.99	37.0%
Well 8	83	22,025.71	27,340.67	24.1%
Well 9	63	9,785.34	11,443.53	16.9%

Key well results. Well 1

- 139% increase in cumulative production
- 225 MCFD average production increase
- 63-day optimization window

63-day optimization window

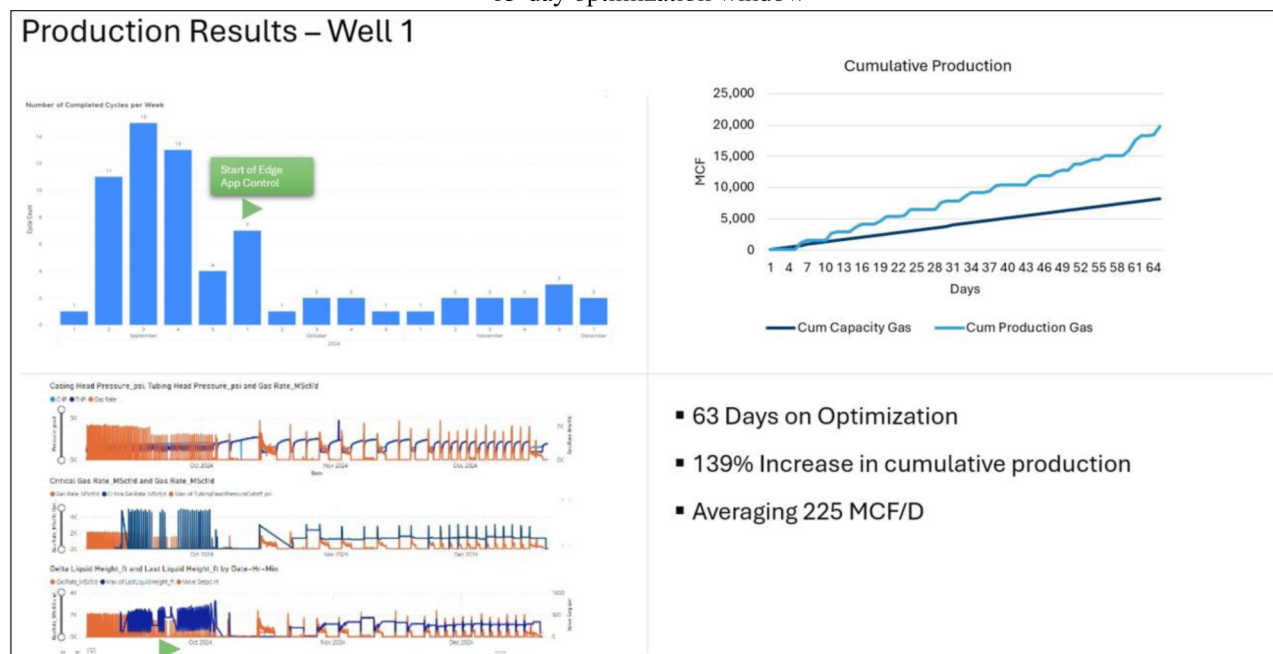


Figure 10—Well 1 results summary card

Findings:

- ML-driven shut-in adjustment: Model extended shut-in durations, reducing cycle frequency and lowering actuator duty, contributing to higher production and improved equipment life.
- Sustained flow above critical velocity: Longer shut-ins enabled gas rates to remain above critical velocity for extended periods, delaying liquid loading and stabilizing cycle efficiency.

Well 2

- 114% increase in cumulative production
- 269 MCFD average
- 83-day optimization window

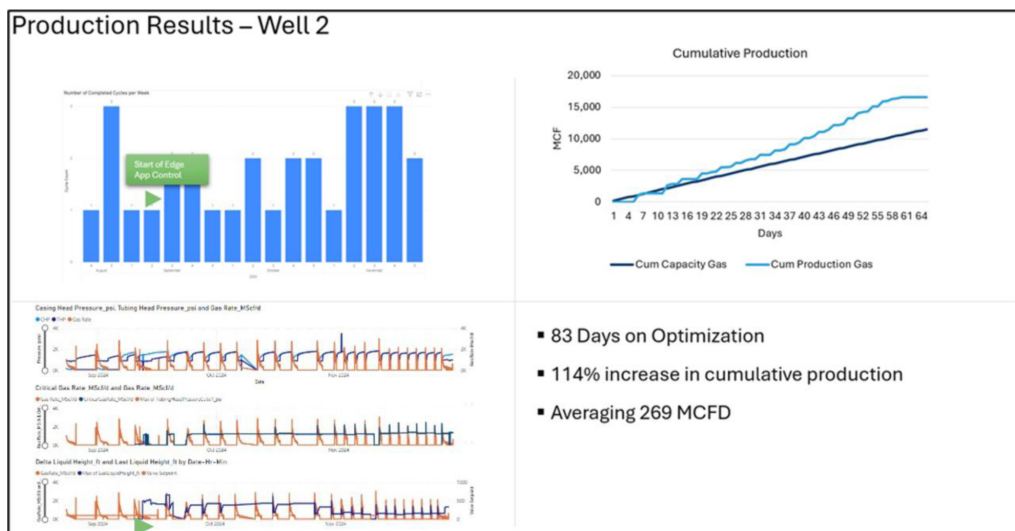


Figure 11—Well 2 results summary card

Findings:

- Early shut-in enforcement: Algorithm consistently closed the well before severe liquid accumulation, preventing inefficient cycles and reducing downtime.
- Optimized shut-in duration: ML-driven adjustment of closed-in period improved pressure recharge efficiency, maximizing flow recovery on reopening.
- Front-loaded production profile: Analysis showed that >50% of total cycle gas output occurred within the first third of the open period, emphasizing the importance of timely shut-in to maximize effective production while minimizing low-rate tail flow.

Well 3

- 70% cumulative production improvement
- Highest uplift in daily average: 350 MCFD
- 83-day optimization period

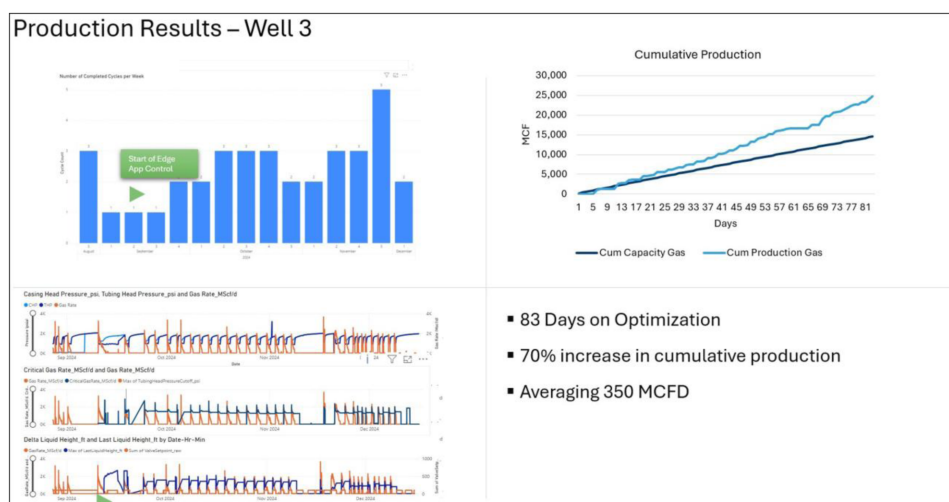


Figure 12—Well 3 results summary card

Findings:

- Consistent with Well 2 and others: More than half of total cycle gas output occurred in the first third of the open period, confirming the same front-loaded production trend seen across the well set.
- Early shut-in enforcement: Algorithm consistently closed the well before liquid accumulation reached critical levels, sustaining efficient cycles.
- Optimized recharge: ML-driven adjustment of shut-in duration improved pressure recovery, supporting stronger flow upon reopening.

Conclusion

This paper presents a field-validated, autonomous control system for optimizing liquid unloading cycles in intermittent gas wells. Deployed in the Haynesville Basin, the solution combines deterministic flow modeling with lightweight machine learning to make intelligent, real-time decisions directly at the well site—without human intervention or reliance on cloud infrastructure.

By continuously evaluating unloading conditions using surface pressure and flow data, the system dynamically controls choke actuation to preempt inefficient cycles and minimize downtime. The integration of a machine learning model for shut-in duration further enhances recovery, adapting to well-specific pressure build-up behavior and flowback performance.

Key outcomes from the field trial include:

- Substantial uplift in cumulative and daily gas production
- Reduction in manual interventions and SCADA dependency
- Rapid deployment with minimal infrastructure changes
- Improved operational consistency across a heterogeneous well population

Importantly, this approach demonstrates how intelligent edge applications can unlock latent value from late-life assets without costly retrofits. As the industry continues its shift toward low-cost, data-driven operations, solutions like this offer a practical, scalable path to improving production, reducing emissions, and extending asset life in maturing basins. Looking ahead, the integration of machine learning into edge-based control systems offers a scalable path to continuous improvement. By retraining models with each new cycle, the system adapts to long-term changes in well performance, reservoir behavior, and equipment dynamics—without requiring reprogramming or manual tuning. This closed-loop learning cycle ensures that optimization becomes progressively more precise over time, even in the face of non-stationary well conditions. As more wells are brought online, the system architecture is designed to support horizontal scaling and model versioning, enabling broader deployment across diverse asset portfolios.

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